

WASHINGTON DULLES, VA, USA

WMO: 724030

Lat: 38.9767N

Lon: 77.4858W

Elev: 88

StdP: 100.27

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93738

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-8.5	-19.8	0.7	-8.8	-17.1	0.8	-6.3	12.2	0.0	11.2	-0.1	3.2	310	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	34.0	23.6	32.6	23.1	31.3	22.6	25.3	31.4	24.6	30.0	24.0	28.9	3.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	18.6	27.4	23.0	17.9	26.8	22.4	17.3	26.4	77.4	31.4	74.5	30.1	72.0	29.0	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.5	7.6	DB	-15.3	36.0	3.0	2.0	-17.5	37.5	-19.2	38.6	-20.9	39.7	-23.0	41.2	(4)
(5)			WB	-15.9	26.7	2.8	1.0	-17.9	27.4	-19.6	28.0	-21.1	28.6	-23.2	29.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	1.0	2.5	6.9	13.0	17.8	22.6	25.0	24.3	20.5	13.9	7.8	3.4	(6)
(7)		DBStd	9.55	5.85	5.38	5.38	4.86	4.33	3.28	2.67	2.73	3.72	4.55	4.74	4.92	(7)
(8)		HDD10.0	979	286	217	128	23	1	0	0	0	0	13	97	214	(8)
(9)		HDD18.3	2532	538	444	356	172	64	6	0	0	21	152	316	464	(9)
(10)		CDD10.0	2178	7	7	32	113	244	379	464	443	314	134	31	9	(10)
(11)		CDD18.3	688	0	0	2	12	49	135	206	185	84	14	1	0	(11)
(12)		CDH23.3	6402	0	1	21	159	468	1220	2014	1699	691	124	4	1	(12)
(13)		CDH26.7	2401	0	0	4	42	141	441	849	671	227	26	0	0	(13)
(14)	Wind	WSAvg	3.0	3.5	3.6	3.8	3.6	3.0	2.7	2.5	2.4	2.6	2.8	3.0	3.1	(14)
(15)	Precipitation	PrecAvg	1042	69	69	87	79	101	102	89	100	95	78	83	81	(15)
(16)		PrecMax	1668	168	159	194	187	261	462	184	272	286	233	199	171	(16)
(17)		PrecMin	733	10	6	25	8	20	13	24	11	16	0	6	11	(17)
(18)		PrecStd	204	38	40	35	41	54	78	43	55	66	56	50	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.6	21.2	26.0	30.7	32.0	34.5	36.1	35.4	33.5	29.4	23.6	20.6	(19)
(20)			MCWB	15.5	14.3	16.1	19.4	21.8	23.8	24.6	24.6	22.2	20.9	16.6	16.0	(20)
(21)		2%	DB	16.6	17.1	22.2	27.6	30.1	32.4	34.2	33.7	31.5	26.6	20.7	16.9	(21)
(22)			MCWB	13.5	11.5	14.2	17.4	21.3	23.3	23.9	23.5	21.8	19.4	15.3	14.1	(22)
(23)		5%	DB	13.4	13.7	19.1	24.8	28.2	30.9	32.7	31.9	29.3	24.1	18.3	13.8	(23)
(24)			MCWB	10.2	8.8	12.6	16.3	20.2	22.5	23.6	23.0	21.0	18.1	14.1	10.6	(24)
(25)		10%	DB	9.5	10.8	16.3	22.1	26.1	29.3	31.2	30.4	27.3	21.9	16.1	11.0	(25)
(26)			MCWB	6.3	6.9	10.9	15.0	19.2	21.8	23.0	22.2	20.2	17.4	12.0	7.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	16.0	17.4	20.8	23.8	25.6	26.4	25.9	24.7	23.0	18.6	17.3	(27)
(28)			MCD B	18.2	20.2	23.5	26.9	29.9	31.8	33.2	32.3	29.5	26.1	21.0	19.3	(28)
(29)		2%	WB	13.9	12.4	15.6	19.1	22.5	24.5	25.4	25.1	23.6	21.3	16.9	14.5	(29)
(30)			MCD B	16.5	15.6	20.5	25.0	28.1	30.3	31.6	31.0	27.8	24.8	19.6	16.8	(30)
(31)		5%	WB	10.3	9.7	13.6	17.5	21.3	23.7	24.7	24.3	22.8	19.8	14.8	11.4	(31)
(32)			MCD B	12.9	13.1	18.1	23.1	26.3	28.9	30.3	29.3	26.6	23.2	17.4	13.3	(32)
(33)		10%	WB	6.8	7.2	11.4	15.9	20.2	22.9	24.0	23.6	21.8	17.8	12.7	8.1	(33)
(34)			MCD B	9.0	10.5	15.8	21.1	24.6	27.5	29.0	28.1	25.4	21.0	15.4	10.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.0	10.0	11.2	12.4	11.5	10.9	10.8	10.9	11.0	11.6	10.9	9.2	(35)
(36)			MCD BR	13.1	14.5	15.8	16.3	14.0	12.6	12.6	12.5	13.4	14.3	14.3	12.7	(36)
(37)			MCWBR	10.3	9.9	9.6	8.1	6.7	5.1	4.4	4.5	5.4	7.6	9.4	9.7	(37)
(38)		5% WB	MCD BR	11.8	12.4	13.9	13.5	12.0	11.0	10.7	10.6	9.9	11.3	10.9	11.0	(38)
(39)			MCWBR	10.1	9.6	9.5	7.6	6.0	5.0	4.4	4.4	4.8	6.8	8.6	9.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.322	0.356	0.402	0.441	0.504	0.520	0.518	0.432	0.362	0.322	0.308	(40)	
(41)		taud	2.470	2.403	2.376	2.273	2.192	2.055	2.031	2.015	2.235	2.432	2.502	2.498	(41)	
(42)		Ebn at Noon	872	901	901	875	844	789	772	760	811	844	845	841	(42)	
(43)		Edh at Noon	80	97	111	131	145	166	169	168	127	94	77	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.04	2.89	3.84	4.85	5.46	6.02	5.90	5.27	4.31	3.23	2.33	1.79	(44)	
(45)		RadStd	0.18	0.33	0.25	0.37	0.58	0.46	0.32	0.27	0.53	0.36	0.20	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.52	N/A	N/A	N/A	N/A	N/A	N/A	-100	+148	+82		
(47) Regional (64 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	+68		

Nomenclature: See separate page